

SIF-Sense AI: Complete User & Operation Guide

A-to-Z Operational Manual: Step-by-Step Walkthrough of All Application Pages, Features, Buttons & Actions

Web Interface: http://localhost:5173	Primary Users: Safety Managers, HSE Officers, Supervisors
Key Workflow: Real-time Incident Analysis & Early Warning	Data Cloud: MongoDB Atlas (kanishkscet_db_user)

1. System Overview: What is SIF-Sense AI?

SIF-Sense AI is an intelligent industrial safety platform designed to identify **Serious Injury or Fatality (SIF) Precursors** before accidents occur. In traditional safety, 80% of minor incidents have zero potential for fatality, while 20% contain latent precursor conditions (such as working at height without harnesses, unverified confined space gas testing, or bypassed lockout tagout). SIF-Sense AI automatically ingests text observations, classifies their severity using a model trained on 105,996 federal OSHA severe incidents, and triggers human-in-the-loop early warning mitigation.

2. Page-by-Page Operational Walkthrough

Page 1: SIF Incident Analyzer (Route: /)

The Analyzer is where users submit, paste, or test safety observations in real time.

- **Dual-Mode Layout:** On desktop, you see the input box on the left and the live safety result on the right. On mobile or narrow viewports ($\leq 860\text{px}$), you see smooth tabs: '1. Describe Incident' and '2. Safety Result'. As soon as you tap 'Analyze Incident', the app automatically flips to Tab 2 to show your result.
- **Quick OSHA Presets:** Tap any of the pre-loaded real OSHA incident buttons (e.g. *Machinery Entanglement*, *Fall from Height*, *Arc Flash Burn*, *Confined Space Gas*) to populate and evaluate real-world cases instantly.
- **Reading the Results:**
 1. **SIF Status Pill:** Glowing RED indicates a confirmed SIF Precursor; glowing GREEN indicates a low-risk routine observation.
 2. **Priority Gauge:** Shows an overall risk score from 0 to 100. Scores ≥ 80 automatically escalate to the Alerts Hub.
 3. **Three Key Takeaways:** Highlights *Predicted Injury Nature* (e.g., Amputation, Fracture, Burn), *Critical Barrier Failure*, and *IOGP Safety Rule*.
 4. **Word Highlights Cloud:** Shows which exact words triggered the high score with their statistical weights (e.g., *amputation* (+21.0), *fingers* (+4.6)).
 5. **Required Action:** Clear plain-English instruction on what site supervisors must do immediately.

Page 2: HSE Command Center Dashboard (Route: /dashboard)

Provides facility-wide intelligence for HSE directors and executives:

- **Top Metric Cards:** Total Reports, SIF Precursors Detected, Critical Incidents, High/Medium Priorities, Near Misses, and Pending Reviews.
- **High-Risk Locations:** Ranks facilities and operating zones by aggregate danger score with proportional progress bars.
- **Hazard Distribution:** Interactive bar chart breaking down recurring hazards (e.g., energized work, falls, rotating equipment).
- **Critical Barrier Breakdown:** Tracks control breakdowns (missing PPE, missing permits, bypassed locks).
- **Sequential Risk Score Trend:** Rolling chronological line chart to spot site safety deterioration before accidents occur.

Page 3: OSHA Model Telemetry & Playground (Route: /model)

A full transparency console explaining how the AI model works and allowing interactive testing:

- **Model Scorecards:** Accuracy (96.00%), SIF Recall (97.49%), Precision (97.18%), F1-Score (0.9733), ROC-AUC (0.9876), and Sub-15ms Latency.
- **Test Set Confusion Matrix:** Displays exact counts of 15,445 True Positives, 4,907 True Negatives, 449 False Positives, and only 398 False Negatives (1.8% missed).
- **Searchable Risk Dictionary:** Interactive dictionary of top risk tokens (fractured +24.5, amputated +22.8, broken +21.9, burns +15.3, concussion +13.6).
- **Live Interactive Playground:** Type any custom text sentence and watch the model calculate probability and highlight words in under 15 milliseconds.

Page 4: Alerts & Human-in-the-Loop Review Hub (Route: /alerts)

When an incident scores ≥ 80 / 100, it automatically triggers an active SIF Early Warning. The Alerts Hub is where safety managers take action:

- **1-Click Direct Actions:**
 - ✓ **Accept & Mitigate:** Acknowledges the hazard and applies recommended controls.
 - **Escalate to Leadership:** Opens a modal to input immediate work stoppage orders, notifications to site leadership, and re-inspection directives.
 - X **Dismiss:** Flags the observation as low-risk or false alarm with audit trail logging.
- **Status Filter Tabs:** Filter between *Pending Action*, *Accepted*, *Escalated*, and *Dismissed*.
- **Dynamic Header Counter:** Topbar displays an active red pill with the number of pending alerts.

Page 5: Safety Report Log (Route: /reports)

Historical incident repository with search, filtering, and export capabilities:

- **Configurable Pagination:** Toggle between 10, 25, or 50 records per page with next/previous controls.
- **1-Click Export to CSV & JSON:** Export data formatted for Excel, PowerBI, or corporate EHS compliance software.
- **Deep Inspection Modal:** Click any report card to open a full inspection drawer showing unedited incident text, priority scores, and extracted precursors.
- **Instant Search & Filter:** Filter specifically for SIF-only incidents or search by facility name, hazard, or activity.

Page 6: Database Cloud Configuration (MongoDB Atlas)

SIF-Sense AI supports multi-user cloud synchronization to MongoDB Atlas with an automatic SQLite fallback.

- **Topbar Button (■ DB Ready):** Click this button in the top navigation bar at any time to open the Database Cloud Modal.
- **Live Status:** Displays your configured user (kanishkskcet_db_user), database name (sif_sense), and connection health.
- **1-Click Test & Connect:** Enter your MongoDB Atlas cluster hostname (e.g. cluster0.abcd.mongodb.net) and click 'Test & Connect' to synchronize immediately without restarting the app.

Quick Reference Checklist for Operators:

1. Enter incident in **Analyzer** → Check SIF status and immediate required action.
2. If Priority ≥ 80 , go to **Alerts Hub** → Click *Accept & Mitigate* or *Escalate*.
3. Review site-wide trends in **Command Center** → Identify deteriorating locations before accidents happen.
4. Export logs in **Report Log** → Download CSV/JSON for enterprise regulatory compliance.